Role of Conformational Entropy in Complex Macromolecular Systems

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Conformation is the key to revealing the physical characteristics of macromolecular systems and receives tremendous interest from the fields of polymer physics and biological materials. The conformational entropy, related to the number of conformations of the macromolecule, plays a predominant role in the structural formation, transition, and dynamics of macromolecular systems. In this review, we present a comprehensive overview of the research, development and applications of the conformational entropy in complex macromolecular systems. We begin by discussing the physical origin of the conformational entropy based on statistical mechanics of macromolecules in classical polymer physics, and then introduce the recent progress on the predictive modeling of the conformational entropy, associated with a variety of typical macromolecular systems. Furthermore, we also highlight several principles and rules, which have been harnessed to manipulate the structural organization of complex macromolecular systems through the conformational entropy. We anticipate that this review will further promote fundamental research in polymer physics, and offer intriguing prospects for applications in complex macromolecular systems including biomacromolecules, grafted nanoparticles, and polymer nanocomposites.

Keywords Polymer conformation; Entropy; Stiffness; External field; Confinement

1 Introduction

Conformation, which denotes the spatial arrangements of the consecutive atoms or atomic groups in a molecule, is of great importance in soft matter physics[1,2], and influences the mechanical[3], thermal[4], electrical[5], and optical[6] properties in macromolecular systems. For a polymer chain composed of thousands of single bonds, the rotational degrees of freedom around its adjacent single bond give rise to its different conformations. In view of statistics, a monomer has n isomers in a polymer chain with N monomers, the total number of the conformations can be estimated as n^N, which takes an exponential amount of N and becomes unlikely to be enumerated. To evaluate the degree of freedom of molecular conformation, the conformational entropy, a thermodynamic amount reflecting the number of conformations that a molecule can take, has been introduced, which enables the perspective of all conformations in macromolecular systems.

Conformational entropy estimates the information of conformations coded in the macromolecules, and governs the structures and morphologies of the macromolecular system. Many important behaviors and processes in material and biological science are involved in the research of conformations in complex macromolecular systems, typically as proteins, grafted nanoparticles, polymer nanocomposites and hydrogels. For instance, the conformational entropy of proteins is the major contributor to the free energy, which governs the structures, activities, binding affinities, and stability of proteins[7−9]. The assembly of the polymer-grafted nanoparticles in polymer matrices exhibits remarkably rich structural morphologies, due to the complex interplay between chain conformational entropy and enthalpic interactions[10−20]. The conformational entropy of the strands in hydrogels can significantly affect the elasticity[21−23], phase separation[24], and diffusion behaviors of the nanoparticle swelling in it[25−28]. There is no doubt that the conformational entropy plays a predominant role in structural formation, transition, and dynamics of macromolecular systems, and has also received increasing interest in rational design of macromolecular materials with tailored structures and robust functionalities.

As the importance of macromolecular conformations is increasingly recognized and their diverse roles in structural formation, transitions, and dynamics are identified, it is crucial to develop approaches that enable us to understand the variables governing the conformational entropy in complex macromolecular systems. Theoretical studies of the conformational entropy have witnessed remarkable progress in classical polymer physics[2,29,30]. However, understanding the role of conformational entropy in complex macromolecular systems remains a challenging task, because these systems exhibit a variety of behaviors that are influenced by different variables like external forces, intermolecular interactions, and environmental conditions. With the development of

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computational techniques, numerical approaches, such as self-consistent field (SCF)[3,24,31−35], Monte Carlo (MC)[11,25,36] and molecular dynamics (MD)[10−20] are widely used in the studies of complex macromolecular systems, providing an alternative means to generate the equilibrated conformations and calculate the thermodynamic quantities, including the free energy, energetic and entropic interactions. Molecular characterization methods like solution nuclear magnetic resonance (NMR) spectroscopy[37], atomic force microscopy (AFM)[38], and dynamic light scattering (DLS)[39] are particularly well suited to measuring conformational dynamics over a wide-range of time scales[40]. Nevertheless, the implementation of conformational entropy in governing the structures, thermodynamics and kinetics still lacks in-depth research, and this field is poised to develop new approaches to tailoring conformational entropy in complex macromolecular systems.

In this review, we present a comprehensive overview of the role of conformational entropy in complex macromolecular systems. We first provide a clear explanation on the physical origin of the conformational entropy, by revisiting the theories of statistical mechanics of macromolecules in classical polymer physics. Recent advances in structure formation, transition, and dynamics of macromolecular systems that involve the implementation of conformational entropy, are thereby introduced. Furthermore, we focus on principles, rules, and strategies, which have been harnessed to manipulate the structural organization of complex macromolecular systems through the conformational entropy. We believe that this review will not only provide a tutorial for physicists, chemists, and biologists to understand the key characteristics featuring conformational entropy in macromolecules, but also expand the field of applications of conformational entropy-mediated strategies in the rational design of new materials.

2 Physical Explanation on the Conformational Entropy in Classical Polymer Physics

Entropy, S, is a basic quantity in thermodynamics, and quantifies the degrees of freedom of a system. Many types of entropy, such as translational entropy, vibrational entropy and conformational entropy are identified to be vital to structural organization, phase transition, and dynamics in polymer and biological systems[14,41−43]. Translational entropy refers to the entropy associated with the translational motion of constituent particles or molecules in a system, accounting for the freedom of movement and randomness in their positions. Vibrational entropy arises from the thermal motions of atoms or molecular vibrations, and represents the freedom of atoms to vibrate. Conformational entropy arises in systems with flexible components, such as polymers or proteins that can adopt multiple conformations. The number of conformational degrees of freedom is specific to each system and is dictated by the number of distinct ways the system can arrange its flexible segments. In this section, we present the basic concept of conformational entropy in classical polymer physics[30,44,45], and discuss the entropic contribution to the free energy and conformations in macromolecular systems. These theories based on coarse-grained models provide useful frameworks for understanding the conformations of macromolecules, and statistically underlying the physical origin of conformational entropy that regulates the molecular architecture in equilibrium states.

We begin with a statistical description of an ideal chain. As the most basic structure in macromolecular systems, the ideal chain, which assumes that no interactions exist except for the bonds responsible for chain connections, becomes a neat theoretical model that is available for describing the statistical behaviors of macromolecules. In the 30s of the last century, it was first proposed for rubber elasticity by Mark and Guth[46−48]. To investigate the visco-elasticity of the polymer solution, Kuhn[49] developed the simplest and perhaps the best known model, the freely jointed chain (FJC) model, underlying the physical origin of the entropic elasticity of an ideal chain with elegant mathematical formation. As shown in Fig.1(A), considering a single polymer chain consisting of N bonds with the same length, b_0, the end-to-end vector of the polymer chain takes the form,

$$\boldsymbol{R} = \Delta\boldsymbol{R}_1 + \Delta\boldsymbol{R}_2 + \cdots + \Delta\boldsymbol{R}_i = b_0 \sum_{i=1}^{N} \boldsymbol{u}_i \tag{1}$$

where $\boldsymbol{u}_i$ denotes the unit vector of the ith bond, and the displacement between the ith and (i+1)th monomers is defined as $\Delta\boldsymbol{R}_i$. Given the random nature of the displacements between consecutive monomers, the mean end-to-end distance, $\langle\boldsymbol{R}\rangle=0$, and the mean squared end-end vector can be written as,

$$\langle\boldsymbol{R}^2\rangle = \langle\boldsymbol{R}\cdot\boldsymbol{R}\rangle = b_0^2 \sum_{i=1}^{N}\sum_{j=1}^{N}\langle\boldsymbol{u}_i\cdot\boldsymbol{u}_j\rangle = b_0^2 \sum_{i=1}^{N}\sum_{j=1}^{N}\langle\cos\theta_{ij}\rangle \tag{2}$$

where θ_{ij} represents the angle between vectors $\boldsymbol{u}_i$ and $\boldsymbol{u}_j$.

The FJC model is a coarse-grained model assuming that there is (1) no bond angle (directional) preference, (2) free rotation at the bond junctions, and (3) no self-interactions (*e.g.*, excluded volume effects) in the absence of external constraints[1]. Using this assumption, the correlation between the different bond vectors is vanished, that is, $\langle\cos\theta_{ij}\rangle = \delta_{ij}$, where δ_{ij} is the Kronecker delta function. Therefore, Eq. (2) can be transformed as $\langle\boldsymbol{R}^2\rangle = Nb_0^2$, which can be regarded as a measure of the spatial extent of the macromolecule.

To study the statistical property of the FJC model, the probability distribution function (PDF) of the end-to-end vector, $P(\boldsymbol{R})$, is employed, which presents the ensemble average of all conformations and becomes the key to obtaining

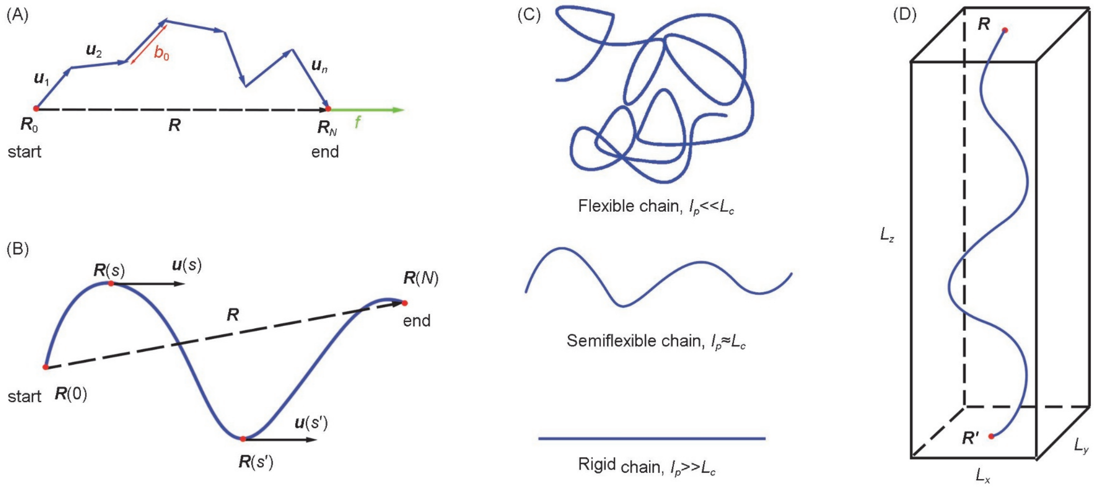

Fig.1 Illustration of theoretical models of ideal chains

(A) Schematic representation of freely-jointed chain (FJC) model; (B) worm-like chain (WLC) model; (C) flexible, semi-flexible and rigid chains; (D) an ideal chain confined in a box.

the conformational entropy. The conformation of an FJC is analogous to the trajectory of a Brownian particle, for which the increment $\Delta \boldsymbol{R}_i$ is independently and identically distributed (IID). Therefore, according to the central limit theorem, if $N \to \infty$, $P(\boldsymbol{R})$ takes the Gaussian form in three-dimensional space,

$$P(\boldsymbol{R}) = \frac{\Omega(\boldsymbol{R})}{\Omega_{\text{tot}}} = \left(\frac{3}{2\pi N b_0^2}\right)^{\frac{3}{2}} \exp\left(-\frac{3\boldsymbol{R}^2}{2Nb_0^2}\right) \tag{3}$$

where $\Omega(\boldsymbol{R})$ represents the number of the macroscopic states of the FJC system. The total configuration partition function of the bond angle $\Omega_{\text{tot}}=(4\pi)^N$, where 4π represents the integral over all possible orientations of a monomer. Then, we turn to obtain the conformation entropy of the FJC model. The conformational entropy S_{conf} can be written as,

$$S_{\text{conf}}(\boldsymbol{R}) = k_{\text{B}} \ln \Omega(\boldsymbol{R}) \tag{4}$$

where k_{B} is the Boltzmann constant. By substituting Eq. (3) into Eq. (4), the conformation entropy as a function of $\boldsymbol{R}$ takes,

$$\begin{aligned} S_{\text{conf}}(\boldsymbol{R}) &= -\frac{3k_{\text{B}}\boldsymbol{R}^2}{2Nb_0^2} + \frac{3}{2}k_{\text{B}} \ln\left(\frac{3}{2\pi N b_0^2}\right) + Nk_{\text{B}}\ln(4\pi) \\ &= -\frac{3k_{\text{B}}\boldsymbol{R}^2}{2Nb_0^2} + S_{\text{conf}}(\boldsymbol{0}) \end{aligned} \tag{5}$$

where the conformational entropy of a polymeric ring $S_{\text{conf}}(\boldsymbol{0}) = (3/2)k_{\text{B}}\ln[3/(2\pi N b_0^2)] + S_{\text{tot}}$ with its end-to-end distance R=0, and $S_{\text{tot}} = Nk_{\text{B}}\ln(4\pi)$ is that of a free ideal chain. The number of conformations decreases with the increase of the end-to-end distance R, resulting in the decrease of the conformational entropy. Since the monomers have no interaction energy, the energy of an FJC (or ideal chain) U is zero, and thus the Helmholtz free energy of the FJC system can be calculated as

$$F(\boldsymbol{R}) = U(\boldsymbol{R}) - TS_{\text{conf}}(\boldsymbol{R}) = \frac{3k_{\text{B}}T}{2Nb_0^2}\boldsymbol{R}^2 - TS_{\text{conf}}(\boldsymbol{0}) \tag{6}$$

where T is the temperature.

3 Effect of Constraints on the Conformational Entropy

In this section, we turn to discuss the effect of constraints, such as the bending stiffness, external field, and confinement on the conformational entropy. Specifically, theoretical results in the section are based on an ideal chain at the coarse-grained level.

3.1 Bending Stiffness

Apart from the flexible chains well described by the FJC model, a series of macromolecules, which favor a straight conformation in equilibrium [Fig.1(C)], are semiflexible or rigid due to the bending stiffness, such as DNA[23,50], carbon nanotubes[51,52] and actin filaments[53,54]. Macromolecules with bending stiffness have the ability to encounter the entropic tendency to crumple up to a random coil[55]. From a physical point of view, the correlation among the bond vectors is responsible for the bending stiffness of these macromolecules.

To investigate the effect of the bending stiffness on the conformation of macromolecules, theoretical implementation of the semiflexible or rigid macromolecules, including freely rotating chain (FRC) and worm-like chain (WLC) models[56], accounts for the presence of the correlation among the bond vectors by setting non-uniformly distributed bond angles between consecutive bonds. In FRC model, if the fixed bond angle is θ, the correlation function between bonds is provided as follow,

$$\langle \boldsymbol{u}_i \cdot \boldsymbol{u}_j \rangle = (\cos\theta)^{|i-j|} \tag{7}$$

For large N, taking Eq. (7) into Eq. (2), the mean-squared end-to-end distance can be estimated as,

$$\langle \boldsymbol{R}^2 \rangle = Nb_0^2 \frac{1+\cos\theta}{1-\cos\theta} \tag{8}$$

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Here, we apply the Flory characteristic ratio C_∞, which is defined as $C_\infty = \langle \boldsymbol{R}^2 \rangle/(Nb^2)$. One can regard the FRC as an equivalent FJC of segment length (also called the Kuhn length) b, with the mapping $b^2 = C_\infty b_0^2$, and $C_\infty = (1+\cos\theta)/(1-\cos\theta)$ in the FRC model.

As indicated in Fig.1(B), a continuous representation of the FRC model can be obtained by taking $N \to \infty$, $b_0 \to 0$, $\theta \to 0$, and ensuring that the contour length L_c=Nb_0 and the persistence length l_p=$2b_0/\theta^2$ are finite. Thus, the correlation function in Eq. (7) can be transformed as,

$$\begin{aligned}\langle \boldsymbol{u}_i \cdot \boldsymbol{u}_j \rangle &= \exp[-|j-i| \ln(\cos\theta)] \\ &\approx \exp\left(-\frac{|j-i|b_0}{l_p}\right) \\ &= \exp\left(-\frac{|s-s'|}{l_p}\right)\end{aligned} \tag{9}$$

where continuous contour parameters $s = ib_0$, $s' = jb_0$. This implementation was first proposed by Kratky and Porod[56], called the worm-like chain (WLC) model. The WLC model provides a quantitative way to quantify the bending stiffness by the persistence length l_p, which measures the exponential decay of the correlation function for the bond vectors along the chain. The bending stiffness can also be expressed by an energetic description,

$$H_{\text{bend}} = \frac{k_{\text{B}}Tl_p}{2}\int \mathrm{d}s \left|\frac{\partial \boldsymbol{u}}{ds}\right|^2 \tag{10}$$

which is widely used for describing biomacromolecules, such as dsDNA[23,57] and actin filaments[50,58−60].

While flexible macromolecules are completely dominated by the conformational entropy, the properties of macromolecules with bending stiffness reflect a competition between the conformational entropy and the bending energy. The ratio of the contour length to the persistence length, L_c/l_p, determines the entropic and energetic contributions to the macromolecular system: when L_c/l_p>>1, the conformational entropy plays a predominate role, resulting in a random-coil configuration of the macromolecule in equilibrium [Fig.1(C)]; in contrast, when L_c/l_p<<1, the bending energy has a major impact on the polymer conformation, which acts as a rod. Notably, the semiflexible macromolecules, where L_c/l_p is almost equal to 1, can induce a more evident conformational-entropy effect, which has been demonstrated in many biological and synthetic macromolecular systems[58−62]. The competition between entropic and energetic effects accounts for many unusual physical properties, such as the non-linear elasticity[58,63] and soft glassy behavior[62,64], inspiring us to take advantage of this conflict to design new materials with unique behaviors and properties.

3.2 External Field

As mentioned above, the free energy of an ideal chain depends quadratically on the magnitude of the end-to-end vector $\boldsymbol{R}$. This indicates that the entropic elasticity of an ideal chain is valid for the Hooke's law. As shown in Fig.1(A), an ideal chain is fixed at a point, with another end pulled by an external force $\mathbf{f}$. Therefore, the partition function of an ideal chain under this external force is,

$$\begin{aligned}Q_f &= \int \mathrm{d}^3\,\boldsymbol{R}\,\frac{1}{(2\pi Nb_0^2/3)^{\frac{3}{2}}}\exp\left(-\frac{3\boldsymbol{R}^2}{2Nb_0^2}\right)\times\exp\left(\frac{\boldsymbol{f}\cdot\boldsymbol{R}}{k_{\text{B}}T}\right) \\ &= \exp\left(\frac{Nb_0^2}{6k_{\text{B}}T}\boldsymbol{f}^2\right)\end{aligned} \tag{11}$$

where the first exponential term is due to the conformational contribution, and the second term is the effect of the external force. The force-extension relation can be calculated by

$$\langle \boldsymbol{R} \rangle = \int \mathrm{d}^3\,\boldsymbol{R}\,\frac{\exp(-\boldsymbol{f}\cdot\boldsymbol{R})/(k_{\text{B}}T)}{Q_f} = \frac{Nb_0^2\boldsymbol{f}}{3k_{\text{B}}T} \tag{12}$$

where $\langle \boldsymbol{R} \rangle$ denotes the averaged end-to-end vector, interpreted as the extension of the ideal chain. The entropy of the system for the unconstrained state is thus,

$$\Delta S(\langle \boldsymbol{R} \rangle) = -\frac{3k_{\text{B}}\langle \boldsymbol{R} \rangle^2}{2Nb^2} \tag{13}$$

which corresponds to the conformational entropy of an ideal chain[Eq. (5)]. The Helmholtz free energy due to the external force can be calculated as,

$$\Delta F = \Delta U - T\Delta S = \frac{3k_{\text{B}}T\langle \boldsymbol{R} \rangle^2}{2Nb^2} \tag{14}$$

This result implies that the mechanical response of an ideal chain to the external force significantly depends on the number of conformations, since there are many more crumpled configurations of the polymer than the straight conformation. Extending the polymer reduces the conformational entropy and may increase the free energy of the macromolecular system.

3.3 Confinement

Confinement, especially the geometric or topological constraint, has become an increasingly popular tool to control the structure and dynamics of the macromolecular systems[10,12]. When macromolecules are confined within small spaces or restricted environments, such as nanoscale pores[65], thin films[66], or surfaces[3,31], their conformational freedom and available conformations can be affected. The spatial confinement can cause chain compression or stretching along specific directions, effectively modifying the preferred conformations of the macromolecules. This alteration in the accessible conformational space influences the conformational entropy.

To clearly show the relationship between conformational entropy and confinement, we present the analytical expressions for the conformational entropy of an ideal chain confined in a cube with hard wall boundary conditions[2,29,30]. Consider a macromolecule confined in a straight cube of volume $V=L_xL_yL_z$, with its lengths L_x, L_y, L_z in x, y and z direction. The number of potential conformations can be obtained from the Green function,

$$G(\boldsymbol{R},\boldsymbol{R}';N)=\int_{\boldsymbol{r}(0)=\boldsymbol{R}'}^{\boldsymbol{r}(N)=\boldsymbol{R}}\mathrm{D}\,\boldsymbol{r}\,\mathrm{e}^{-H(\boldsymbol{r},N)} \quad (15)$$

where $\boldsymbol{R}$ and $\boldsymbol{R}'$ represent respectively the start and end points of the macromolecule, and D$\boldsymbol{r}$ is the path integral of the vector $\boldsymbol{r}$. The Hamiltonian of the confined macromolecule is thus,

$$H(\boldsymbol{r},N)=\frac{3}{2b^2}\int_0^N\left(\frac{\partial\boldsymbol{r}}{\partial s}\right)^2\mathrm{d}s+\int_0^N A(\boldsymbol{r},s)\mathrm{d}s \quad (16)$$

where the first term denotes the conformational entropy, and the second term represents a hard potential wall, which can be written as,

$$A(\boldsymbol{r},s)=\begin{cases}0 & \text{if } \boldsymbol{r} \text{ in tube}\\ \infty & \text{else}\end{cases} \quad (17)$$

The solution of the Green's function is equivalent to solving a partial differential equation (PDE),

$$\left(\frac{\partial}{\partial N}-\frac{b^2}{6}\frac{\partial}{\partial\boldsymbol{R}^2}-A(\boldsymbol{R},N)\right)G(\boldsymbol{R},\boldsymbol{R}';N)=\delta(\boldsymbol{R}-\boldsymbol{R}')\delta(N) \quad (18)$$

Noting that the solution of Eq.(18) is factorizable,

$$G(\boldsymbol{R},\boldsymbol{R}';N)=G(x,x';N)G(y,y';N)G(z,z';N) \quad (19)$$

with the general solution using the eigenfunction expansion method[29],

$$G(\alpha,\alpha';N)=\frac{2}{\alpha}\sum_{p_\alpha=1}^{\infty}\sin\left(\frac{\pi\alpha}{L_\alpha}p_\alpha\right)\sin\left(\frac{\pi\alpha'}{L_\alpha}p_\alpha\right)\exp\left(-\frac{p_\alpha^2\pi^2Nb^2}{6L_\alpha^2}\right) \quad (20)$$

As shown in Fig.1(D), we set the start point at $\boldsymbol{R}$=(L_x/2, L_y/2, 0) and the end point at $\boldsymbol{R}'$=(L_x/2, L_y/2, L_z). Taking the first-order approximation of Eq. (20) in the x and y directions, and substituting into Eq. (19), the Green function can be simplified as,

$$G(\boldsymbol{R},\boldsymbol{R}';N)=\frac{4}{L_xL_y}\exp\left(-\frac{\pi Nb^2}{3L_xL_y}\right)\left(\frac{3}{2\pi Nb}\right)^{\frac{1}{2}}\exp\left(-\frac{3L_z^2}{2Nb^2}\right) \quad (21)$$

To calculate the conformational entropy of the confined chain, we evaluate its potential conformations proportional to those of a free ideal chain,

$$G(\boldsymbol{R},\boldsymbol{R}';N)=\frac{\Omega_{\text{confine}}(\boldsymbol{R},\boldsymbol{R}';N)}{\Omega_{\text{tot}}} \quad (22)$$

and the corresponding conformational entropy can be realized as,

$$S_{\text{confine}}=k_{\mathrm{B}}\ln\Omega_{\text{confine}}\approx S_{\text{conf}}(\boldsymbol{0})-\left(\frac{\pi Nb^2L_z}{3V}+\frac{3L_z^2}{2Nb^2}\right) \quad (23)$$

As indicated in Eq.(23), macromolecules experience a reduction in conformational entropy when squeezed into a small space, such as a narrow slit between two parallel plates or a straight narrow tube, because fewer conformations are available to the repeat units in confinement than in free space. Taking the unlimited condition $V\to\infty$, Eq. (23) can be simplified to the result of an unconfined ideal chain in Eq. (5). These analytical results have demonstrated that the confinement can have a great impact on the conformational entropy, which allows us to develop entropic strategies by altering the confinement degree of the macromolecule.

The statistical mechanism of the external constraints that directly regulates the conformational entropy has been revealed through theoretical analysis. We can expect to see analytical solutions to the conformational entropy that are based on solid foundations in polymer physics. However, characterizing the effect of conformational entropy on the complex macromolecular systems would be substantially tough, due to the multiple interactions between structural units in macromolecules. This frustrates the pursuit of the rules and principles, which have been harnessed to manipulate the structural organization of complex macromolecular systems through the conformational entropy.

4 Implementation of Conformational Entropy in Complex Macromolecular Systems

In this section, we will review the recent studies on the role of the conformational entropy in complex macromolecular systems, containing block copolymers, polymer nanocomposites, grafted nanoparticles, and biomacromolecular systems, aiming at exploring fundamental principles and rules that regulate the structural organization, phase transition, or dynamics in those systems.

4.1 Block Copolymer (BCP) System

Block copolymers, which can conduct phase separation and self-assemble into ordered structures at the nanoscale, are ubiquitous and provide potential or practical applications in many fields, such as drug delivery[67], nanoreactors[68], self-healing materials[3,69] and track-etching devices[70]. One of the important issues in the self-assembly of the block copolymer is how to precisely control diverse morphologies as well as the domain sizes[71]. Bates and coworkers examined the order-disorder transition (ODT) of A-B diblock copolymer [Fig.2(A)] using self-consistent field theory (SCFT)[33,35] and transmission electron microscope (TEM) [Fig.2(B)][72], suggesting that the phase behavior of BCP was determined by three major quantities: the degree of polymerization N, the composition f_A and the A-B block-block interaction parameters c [Fig.2(C)]. Among them, the first two parameters have an impact on the conformational entropy, which plays a crucial role in the selection of various phases[34].

Theoretical studies have suggested that there is thermodynamic competition between block-block interactions and conformational entropy in block copolymer systems, and the entropic and enthalpic contributions to the free energy of BCP systems scale as N^{-1} and c[73] . These results highlight the influence of conformational entropy due to the chain stretching on its equilibrium structure, even though SCFT

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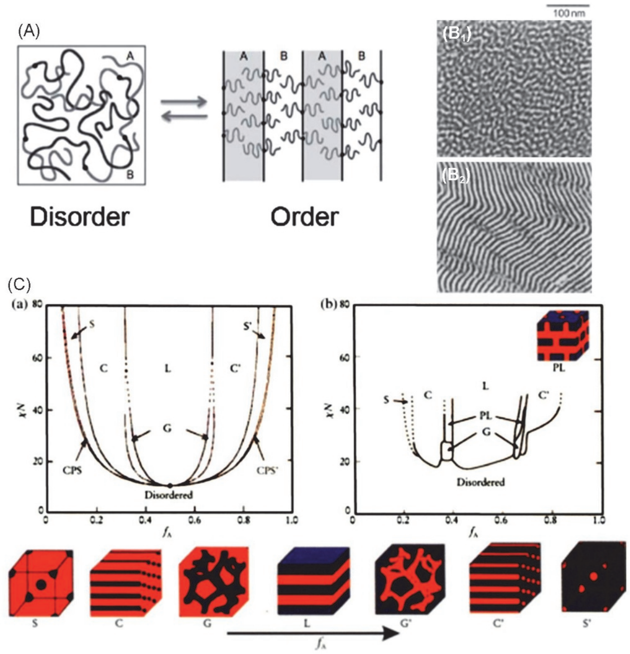

Fig.2 Entropy-mediated order-disorder transition of A-B diblock copolymer

(A) Schematic illustration of chain conformations in disordered state (left) and ordered state (right) of an A–B diblock copolymer; (B_1, B_2) TEM images of disordered state (B_1) and ordered state (B_2) of symmetric poly(styrene-block-isoprene). Reprinted with permission from Ref.[72], Copyright 2015, Woodhead Publishing; (C) phase diagram of diblock copolymer predicted by SCFT. Along the bottom of the figure are representations of four different equilibrium morphologies: S=body-centered-cubic (BCC) spheres, C=cylinders, G=gyroid, and L=lamellae. Other structures found theoretically or experimentally include close-packed spheres (CPS) and perforated lamellae (PL). Reprinted with permission from Ref.[33], Copyright 1999, AIP Publishing.

calculations as a mean-field method, ignore the fluctuations and long-range correlation. This greatly extends the range of applications of the block copolymer materials with predictable morphologies in equilibrium states.

4.2 Polymer Nanocomposites System

Polymer nanocomposites have witnessed more interest due to the enhanced mechanical properties reinforced by mixing nanoparticles within a polymer matrix[74,75]. Controlling the spatial distribution of the nanoparticles inside the polymer matrix is particularly intriguing, since it is essential to harness their full potential and achieve desired material properties[3,76,77]. Tremendous progress has been achieved in manipulating the arrangement of nanoparticles in polymer matrices. In the case of block copolymer nanocomposites, the field theories have been applied to predicting the mesophase of block copolymer and the distribution of nanoparticles in the preferential domain. Balazs and coworkers[32] proposed a combined approach of SCFT and density profile theory (DFT), to incorporate the particle-polymer interaction in SCFT calculations, which provided a possibility for the spatial distribution of polymer segments and nanoparticles. For a large particle with the radius $R=0.3R_0$ (R_0 is the root-mean-square end-to-end distance of block copolymer) and a high particle volume fraction $\varphi_P=0.15$, nanoparticles were excluded by the polymer chains, and segregated in the middle of the preferential domain, while small nanoparticles with $R=0.2R_0$ and $\varphi_P=0.15$ were uniformly dispersed in the domain [Fig.3(A)]. The inclusion of nanoparticles in a polymer matrix can impose steric constraints on the macromolecular chains, leading to reduced conformational entropy. Therefore, more conformational entropy gained by the block chains could compensate for the loss of the translational entropy associated with the aggregation of large nanoparticles. However, this penalty was absent for the smaller nanoparticles and their dispersion inside the domain was favored by an increased translational entropy [Fig.3(B)]. These results have been verified by both experiments [Fig.3(C)][78] and Monte-Carlo simulations [Fig.3(D)][36] in mixtures of block copolymers, homopolymers, and nanoparticles, demonstrating that block copolymers can act as templates for the assembly of nanoparticles into hierarchical structures with tunable spatial distribution.

In addition, the externally applied field can significantly change the hierarchical assembly of nanoparticles in polymer nanocomposites. For instance, by combining molecular dynamics (MD) and Monte Carlo simulations, the distribution of nanoparticles can be tuned by the gap between two polymer brush layers [Fig.3(E)][79]. As the external pressure increased, nanoparticles were gradually immersed into the brush layers, reducing the loss of conformational entropy of grafted polymers under external pressure. Similar behavior is triggered in block copolymer systems, where large nanoparticles transit from the center of the preferential domain to the interface under external pressure [Fig.3(F)][13], in sharp contrast to the equilibrated conformations in Fig.3(A). Theoretical analysis indicates that compared to the compacted conformations in equilibrium, the chains in the preferential domain tend to wrap nanoparticles to gain more conformational entropy under external pressure, resulting in the transition of the nanoparticle distribution. These findings substantiate the importance of the conformational entropy for the self-assembly of polymer-nanoparticle hybrid systems, and provide new insights into the development of polymeric stimuli-responsive materials.

4.3 Polymer-grafted Nanoparticle System

Polymer-grafted nanoparticles, which can exhibit rich phase behaviors and control nanoparticle distribution in composites, have attracted extensive research interest due to their potential for enhancing mechanical and optical properties[18,80]. The self-assembly structure of polymer-grafted nanoparticles is a

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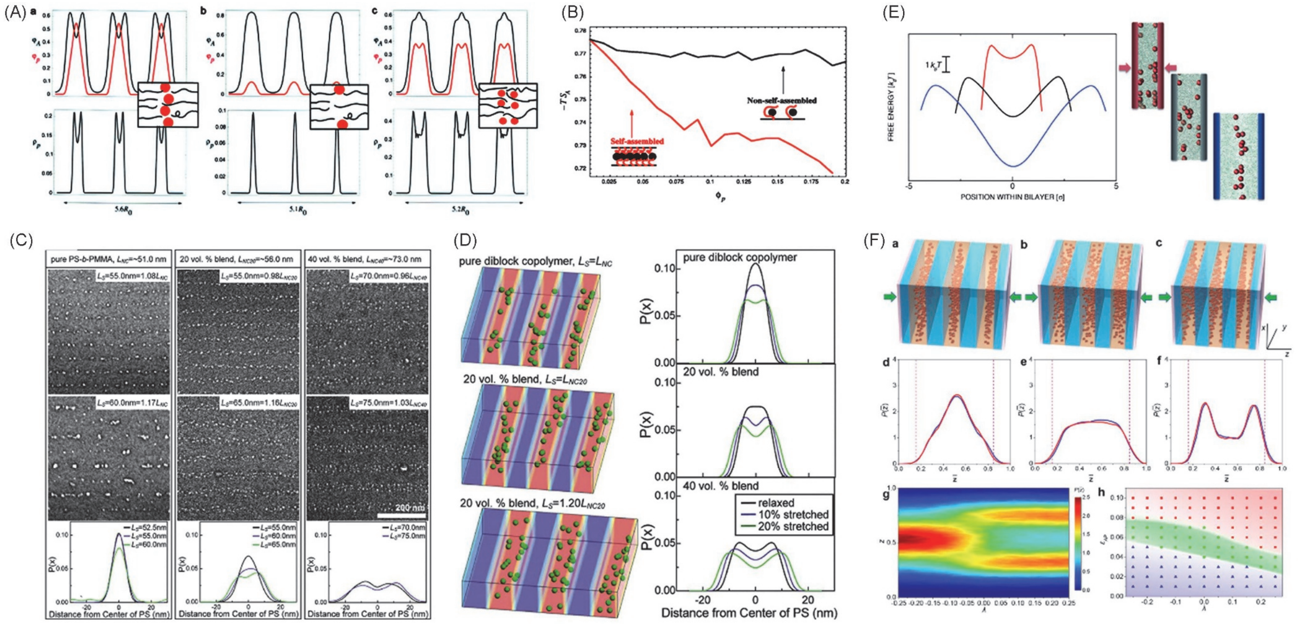

Fig.3 Conformational entropy effect on polymer nanocomposites

(A) Concentration profiles of the nanoparticles (φ_P) in the A block (φ_A) of a diblock copolymer along three periods of a lamellar structure (top row) as well as the distribution of particle centers (ρ_P, bottom row) in units of R_0. a and b. While large nanoparticles are segregated to the center of the domain; c. small nanoparticles are dispersed inside the domain; (B) Entropic contribution free energy of A-block $-TS_A$ as a function of particle volume fraction φ_P. Red and black curves indicate the large and small nanoparticles, respectively. (A) and (B) Reprinted with permission from Ref.[32], Copyright 2001, American Association for the Advancement of Science. (C) Experimental results of nanoparticle-patterning with block copolymers. Reprinted with permission from Ref.[78], Copyright 2008, American Physical Society. (D) MC simulation of the nanoparticle patterning. Reprinted with permission from Ref.[36], Copyright 2008, American Chemical Society. (E) Free energy landscape and snapshots of the nanoparticle between polymer brush bilayers. Reprinted with permission from Ref.[79], Copyright 2014, American Chemical Society. (F) Hierarchical self-assembly of nanoparticles in diblock copolymer nanocomposites under external pressure. a—c. Diagrams showing mechanoresponsive spatial transition of nanoparticles inside their preferential domain; d—f. distribution of nanoparticles inside their preferential domain upon various compression ratios λ, where λ: d. −0.25, e. 0, and f. 0.25; g. evolution of nanoparticle density profile $P(\bar{z})$ with λ increasing from −0.25 to 0.25; h. phase diagram describing the states of the spatial distribution of nanoparticles as a function of λ and ε_{AP}. Reprinted with permission from Ref.[13], Copyright 2019, American Chemical Society.

complex interplay among chain conformational entropy, depletion effects, and enthalpic interactions[14], the former of which depends on properties of grafted chains, including length[15,81], asymmetry[15], grafting density[82] and stiffness[16], and external constraints, such as applied forces[19] and confinement[12]. In the case of the phase separation of Janus nanoparticles at interfaces, the mixing parameter φ, which reflects the dispersity of the nanoparticle-mixing state, is highly correlated with the contour length L_c and persistence length l_p of the grafted chains [Fig.4(A)][17]. The flexible chains tend to collapse in response to interchain repulsion. In contrast, rod-like chains protrude from the nanoparticle surface and exhibit a relatively slight orientation in the direction perpendicular to the interface, leading to structural remixing of Janus nanoparticles. For the semiflexible chains, interchain repulsion can induce a stretched and orientated conformation, which contributes to the large loss of entropy that is required by entropic templating. Experiments, simulations and theoretical analysis show that the conformational entropy of long tethers plays a key role in the mixing process [Fig.4(B)], as the reduction of the available conformations of chains can impose an entropic repulsion between nanoparticles at their approach[38]. Moreover, with the external force applied to the lateral surface, the self-assembly of Janus nanoparticles at interfaces exhibited a mechanical response characterized by a reversible transition between uniform mixing and long-range phase separation [Fig.4(C)][19]. As the lateral pressure is exerted on the systems, the grafted chains repel each other, forcing them to extend perpendicular to the interface. Therefore, the Janus nanoparticle system favors an intercalated state rather than the randomly mixing state to reduce the loss of the conformational entropy.

Another common strategy to dictate the phase separation of grafted nanoparticles is confinement. Confining polymer-grafted nanoparticles in space with various geometries can make a significant difference in the balance between the conformational entropy and enthalpic interactions of grafted chains[66], which have been utilized for endowing unique phase behaviors with remarkable new nanopatterns. For example, using molecular dynamics simulation, Zhu *et al.*[12] investigated the self-assembly of polymer-grafted Janus nanoparticles on a sphere. By systematically changing the length of A-type tethers (L_A) and the surface fraction of A-type tethers (ϕ_A), these nanoparticles could assemble into diverse novel structures, including binary nanoclusters (C_B), trinary nanoclusters (C_T), nanoribbons (R_N), and hexagons with centered reversal (H_R) [Fig.4(D)], which were demonstrated to be the interplay among shape anisotropy of Janus particles, the

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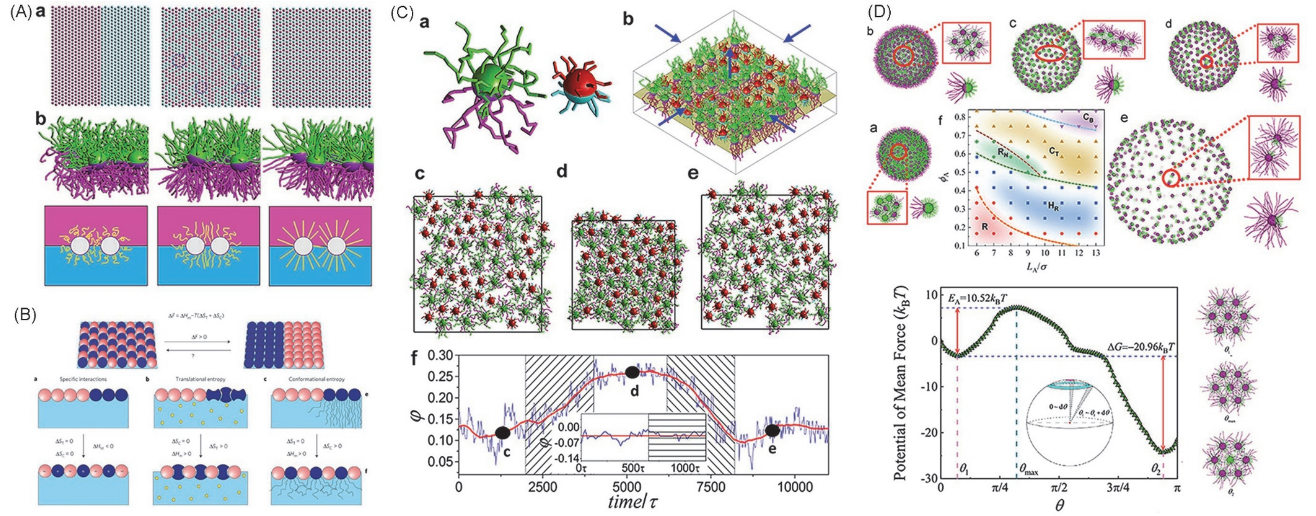

Fig.4 Conformational entropy-meadiated behaviors of grafted nanoparticles

(A) Chain stiffness regulates entropy-templated perfect mixing at the single-nanoparticle level. a. Representative patterns of phase-separated (left), randomly mixing (middle) and intercalated (right) states, where the mixing parameters are –0.33, 0, and 0.33, respectively; b. snapshots (upper) and schematic plots (bottom) showing the conformational transition of the Janus nanoparticle systems for various tethered chains, where l_p=0.50r_c (left), l_p=3.28r_c (middle), and l_p=5.90r_c (right), and r_c denotes the length unit of the system. Reprinted with permission from Ref.[17], Copyright 2015, American Chemical Society. (B) The interplay of specific interactions (a), translational entropy (b) and conformational entropy (c) in self-assembly of grafted nanoparticles at the fluid–fluid interface. Reprinted with permission from Ref.[38], Copyright 2013, Springer Nature. (C) Mechanical response of binary mixtures of grafted Janus nanoparticles at thefluid–fluid interface. Schematics of a Janus nanoparticle tethered with long and short ligand chains (a) and the random arrangement of nanoparticles at the fluid interface for the initial simulations(b); c—e. top-view snapshots of the interfacial nanoparticle patterns at the times denoted by the black points in f.; f. the time dependence of the mixing parameters φ during a successive compressing-stretching process. Reprinted with permission from Ref.[19], Copyright 2014, American Chemical Society. (D) The effects of conformational entropy on the polymer-tethered particles modified by confining them in the spaces with various geometries. a—e. Representative states formed by the self-assembly of tethered Janus nanoparticles with various anisotropies on spherical surfaces. a. Random state (R); b. hexagon with centered reverse (H_R); c. nanoribbon (R_N); d. Trinary nanocluster (C_T); e. binary nanocluster (C_B); f. phase diagram of the structural states on the f_A-L_A plane; g. potential mean force (PMF) as a function of polar angle θ during a reverse of a central nanoparticle in an H_R unit. Reprinted with permission from Ref.[12], Copyright 2020, Elsevier.

curvature of the surface, and the chain conformation of grafted chains. Theoretical analysis revealed that the formation of these hierarchical structures was determined by the free volume and the conformational entropy of grafted chains, indicating a defect-enhanced entropy effect on a curved surface. These results stimulate efforts to develop the entropy-mediated strategies to alter the supramolecular structures using anisotropic polymer-grafted nanoparticles and curved surfaces.

4.4 Biomacromolecular System

Biomacromolecules, such as peptides, proteins, and nucleic acids, are widely spread in life, which can serve as structural and functional units of living things, and become fundamental for biological processes, such as mass transportation, signal transduction, and pathogen transmission[42,83,84]. It has been recognized that the conformational entropy is indispensable for efficiently intercepting the physiologic function of biomacromolecules *in vivo*. Protein molecules can form diverse and complex structures (primary, secondary, tertiary, and quaternary structures) after folding or deformation, which is a necessary condition for the realization of different functions and properties of proteins[37]. The formation of unique spatial structures is of great significance in biology, and conformational entropy is one of the important factors determining the three-dimensional conformation of protein molecules. For example, for protein molecules that have not undergone folding due to stretching, the solvent molecules in their environment will spontaneously and regularly arrange near the hydrophobic amino acid side chains of the protein, resulting in a decrease in the entropy value of water. To avoid a decrease in the total entropy of the reaction system, protein molecules choose to wrap hydrophobic side chains in their three-dimensional structure (including non-polar side chains, polar side chains, peptide groups, charged side groups, etc.) through bending and folding. During the process of stretching and folding, protein molecules exhibit a more regular and orderly three-dimensional structure, losing a significant amount of conformational entropy [Fig.5(A)]. However, the solvent molecules originally arranged around hydrophobic groups lose their constraints and randomly disperse in the solution[8,9]. As the free volume increases, the translational entropy of the solvent molecules significantly increases, resulting in an increase in the total entropy of the protein system in folded conformation.

Recent works have shown that the curvature of cell membranes can be vitally affected by certain special proteins due to the conformational entropy[85−87]. For example, the multifunctional protein Matrix 2 (M2) can induce the bending

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curvature of the membrane, helping the budding and separation processes of the Influenza A virus and promoting its release from the cell, which is of great significance to virus replication [Fig.5(B)][85]. Experiments showed that during the budding and separation processes, M2 protein tended to gather at the neck of the budding body, leading to the generation of a negative Gaussian curvature of the membrane at the budding position[86]. Meanwhile, the amphiphilic helical structure of the tail of the M2 protein is extremely essential for the induction of the negative Gaussian curvature of the membrane[87]. If the hydrophobicity of this helical structure is weakened, the negative Gaussian curvature will be difficult to generate, inhibiting the budding and release processes of the influenza virus. The results amply demonstrate that the change in protein composition and structure will have a huge impact on its conformational entropy, which might be a feasible strategy for designing new antiviral drugs by monitoring the conformational entropy of proteins to inhibit the process of virus replication.

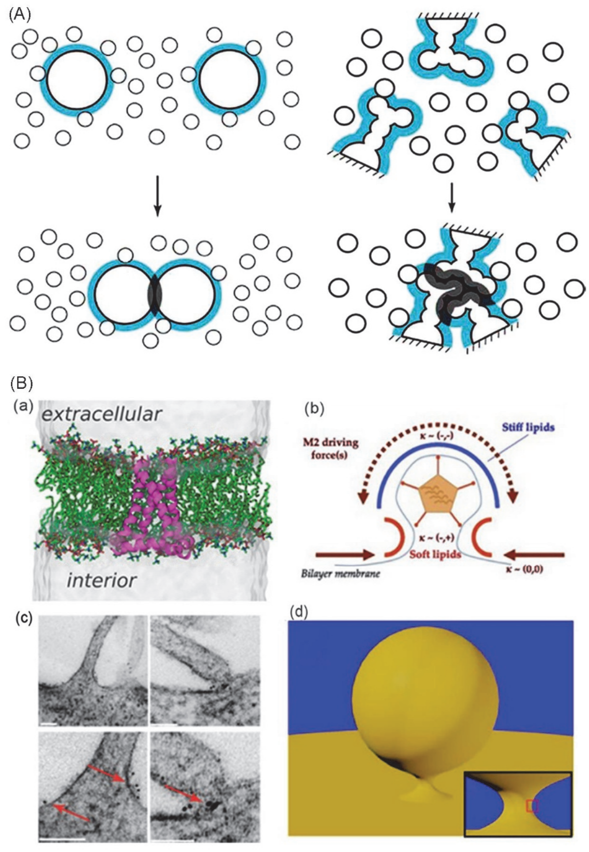

Fig.5 Conformational entropy in biomacromolecular systems

(A) The interplay of translational and conformational entropies on the protein folding process. Reprinted with permission from Ref.[9], Copyright 2004, Wiley. (B) Conformational changes and entropic effect of M2 protein during virus budding process. a.The transmembrane M2 protein in an all-atom simulation of a model biological membrane; b schematic drawing of the proposed model the role of M2 in influenza budding. (a) and (b) Reprinted with permission from Ref.[85], Copyright 2018, National Academy of Sciences. (c) Virus-infected cells stained for M2 *via* immuno-gold labeling and thin sections analyzed by electron microscopy. Arrows indicate M2 foci at sites of virus budding. Reprinted with permission from Ref.[86], Copyright 2010, Elsevier. (d) Illustration of the budding process just before membrane scission, where the Influenza A virus was approximated as a sphere. Reprinted with permission from Ref.[87], Copyright 2013, American Chemical Society.

5 Conclusions and Perspectives

In summary, we have attempted to provide a coherent view of essential concepts of the conformational entropy from the aspect of statistical mechanics in polymer physics. Through theoretical analysis, we conclude three major external constraints that have a significant impact on the conformational entropy, and discuss in detail how they work on the thermodynamic quantities in macromolecular systems. Furthermore, to achieve the full potential of the conformational entropy in tailoring the structural organization, phase transition and dynamics of complex macromolecular systems, we review recent advances in experiments, simulations and theoretical analysis of the conformational entropy, providing a new insight into the fundamental principles and rules for precisely programming the nanostructure through conformational entropy, and thereby gaining the optimal properties of macromolecular materials.

Despite the remarkable progress described above, this field is also plagued by certain challenges. For instance, conformational changes that increase the conformational entropy of a molecule often involve the redistribution of energy within the system. This can lead to an enthalpic cost, where the gain in conformational entropy is compensated by a decrease in enthalpy. The balance between entropy and enthalpy poses a challenge in predicting and understanding the thermodynamics of conformational transitions in complex macromolecular systems. Nonetheless, it can be expected that this review will help readers understand the physical origin of the conformational entropy, and stimulate fundamental research on the conformational entropy-mediated strategy that governs the morphologies and dynamics in macromolecular systems.

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Conflicts of Interest

YAN Li-Tang is a youth executive editorial board member for Chemical Research in Chinese Universities and was not involved in the editorial review or the decision to publish this article. The authors declare no conflicts of interest.

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